

Mid-Term Exam for 《Calculus》

2022-2023 2nd Semester

Class: _____ Name: _____ Sit NO.: _____ Total Score: _____

Note: 1. Please clearly write down the private information needed before exam.

2. Please clearly write down the answers on the paper or answer sheet.

3. Full mark: 100

4. Duration of Exam: 120 minutes

1. Please fill the correct answers in the following blanks. (18 marks)

(1) If $\vec{a} = \langle -1, 2, 2 \rangle$, $\vec{b} = \langle 2, -1, 2 \rangle$, $\vec{c} = \langle 1, 0, 0 \rangle$, then $\vec{a} \times \vec{b} \cdot \vec{c}$ is _____;

(2) Equation $xyz + \sqrt{x^2 + y^2 + z^2} = \sqrt{2}$ defines an implicit function $z = f(x, y)$, the differential of the dependent variable dz at point $(1, 0, -1)$ is _____;

(3) Evaluate iterated integrals:

$$\int_{\frac{1}{4}}^{\frac{1}{2}} dy \int_{\frac{1}{2}}^{\sqrt{y}} e^{\frac{y}{x}} dx + \int_{\frac{1}{2}}^1 dy \int_y^{\sqrt{y}} e^{\frac{y}{x}} dx = \underline{\hspace{2cm}};$$

(4) The equation of the tangent line to the curve $\begin{cases} z = x^2 + y^2 \\ x^2 + 2y^2 + 3z^2 = 8 \end{cases}$ at point $P(0, \frac{2\sqrt{3}}{3}, \frac{4}{3})$ is _____;

(5) The radius of convergence of $\sum_{n=1}^{\infty} \frac{n}{2^n} x^{2n}$ is _____;

(6) Suppose $f(x, y)$ attains its local extreme value at (x_0, y_0) . If $f(x, y)$ is differentiable at (x_0, y_0) , then the directional derivative of f at (x_0, y_0) in arbitrary direction is _____.

2. (6 marks) Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^2 y^2 + (x-y)^2}$ does not exist.

3. (10 marks) Let $f(x, y) = \begin{cases} (x+y)^2 \sin \frac{1}{x^2 + y^2}, & x^2 + y^2 \neq 0 \\ 0, & x^2 + y^2 = 0 \end{cases}$

(1) Find $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$;

(2) Determine the continuity of $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ at the origin;

(3) Determine the differentiability of $f(x, y)$ at the origin. Please show the reason.

4.(6 marks) Let $\xi = \frac{y}{x}$, $\eta = y$, the second partial derivatives of u are continuous. Try to prove the equation $x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} = 0$ can be transformed into the equation $\eta^2 u_{\eta\eta} = 0$.

5.(6 marks) Find the gradient of function $z = x^2 - xy + y^2$ at point $(-1,1)$. In what direction does $z = x^2 - xy + y^2$ decrease most rapidly? And in what direction does $z = x^2 - xy + y^2$ stay at the same level?

6.(6 marks) Find the equations of the tangent plane and the normal line to the given surface

$e^{\frac{x}{z}} + e^{\frac{y}{z}} = 3$ at the indicated point $(\ln 2, 0, 1)$.

7. (6 marks) Find the minimum distance between the origin and the surface $x^2y - z^2 + 9 = 0$.

8.(6 marks) Test for the convergence or divergence of $\sum_{n=1}^{\infty} \frac{n}{n5^n + 2}$.

9.(6 marks) Find the convergence set and the sum function of the power series

$$\sum_{n=1}^{\infty} (-1)^n \frac{(x+2)^{2n}}{n \cdot 3^{n+1}}.$$

10.(6 marks) Expand the function $f(x) = \ln \frac{1+x}{1-x}$ into power series of x .

11.(6 marks) Let $f(x) = \begin{cases} e^{-1/x^2}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$, find the Maclaurin series for $f(x)$. Does the Maclaurin series represent $f(x)$? Why?

12.(6 marks) Find $\iint_D \frac{y}{x+y} e^{(x+y)^2} dx dy$, D is bounded by $x+y=1$, $x=0$ and $y=0$.

13.(6 marks) Find the volume of the solid bounded by the surface $z=xy$ and planes $z=x+y$,
 $x+y=1$, $x=0$, $y=0$.

14. (6 marks) Calculate $\iiint_{\Omega} (x^2 + y^2) dV$, Ω is bounded by the planes $z = 2, z = 8$ and the surface that results when the parabola $y^2 = 2z$ in the yz -plane is revolved about the z -axis.